Roll No. ....

TEC–301

B. TECH. (ECE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

ELECTRONIC DEVICES AND CIRCUITS

Time : Three Hours

Maximum Marks : 100

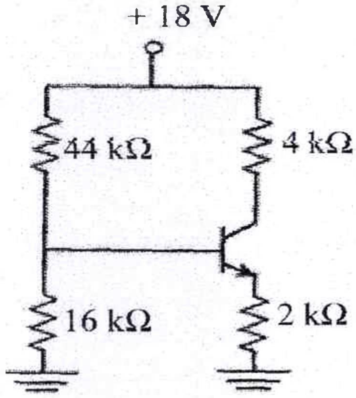
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Why BJT is referred as a current controlled device ? Consider the circuit shown in the figure. Assume base-to-emitter voltage $V_{BE} = 0.8$ V and common base current gain (α) of the transistor is unity. (CO1)



Calculate the value of the collector to-emitter voltage V_{CE} (in volts).

P. T. O.

(b) For a BJT the common base current gain $\alpha = 0.98$ and the collector base junction reverse bias saturation current $I_{CO} = 0.6\ \mu A$. This BJT is connected in the common emitter mode and operated in the active region with a base drive current $I_B = 20\ \mu A$. Calculate the collector current I_C. (CO1)

Explain the following terms : (CO1)

(i) I_{CBO}

(ii) I_{CEO}

(c) Explain base-width modulation and its consequences in detail. (CO1)

2. (a) Explain stability in a silicon transistor circuit with a fixed bias, $V_{CC} = 9$ V, $R_C = 3$ kΩ, $R_B = 8$ kΩ, $\beta = 50$, $V_{BE} = 0.7$ V. Find the operating point and stability factor. (CO1, CO2)

(b) (i) Discuss channel length modulation in MOSFETs, along with a proper schematic diagram. (CO1, CO2)

(ii) Show the dependence of channel length modulation parameter (λ) on drain current. (CO1, CO2)

(c) Explain the frequency response of RC coupled common emitter amplifier. (CO1, CO2)

3. (a) Draw the circuit diagram of RC coupled common source amplifier and derive the equation for voltage gain, input resistance and output resistance. (CO3, CO5)

(b) Explain the significance of threshold voltage of a MOSFET. Discuss the methods to reduce threshold voltage (V_T). (CO3, CO5)

(c) Discuss the transfer and output characteristics of *n*-channel depletion MOSFET.

4. (a) Explain with a neat circuit diagram, the working of Wien Bridge oscillator. Derive an expression for frequency of oscillation.

(CO3, CO4)

(b) Draw the circuit of Hartley oscillator and explain its working. Derive an expression for frequency of oscillation. (CO3, CO4)

(c) (i) Explain the gate capacitive effect in MOSFET. (CO3, CO4)

(ii) Explain the small signal high-frequency model for MOSFET.

(CO3, CO4)

5. (a) Describe the working of transfer coupled Class A power amplifier in detail with a relevant diagram. Derive the expression for conversion efficiency and figure of merit (FOM). (CO5, CO6)

(b) Describe the working of Class B push pull power amplifier. Derive the expression for conversion efficiency and figure of merit (FOM).

(CO5, CO6)

(c) Draw the circuit diagram of common-drain MOSFET amplifier. Derive the expression for its voltage gain, input resistance and output resistance. (CO5, CO6)

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TEC–302

B. TECH. (ECE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

DIGITAL ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Design the circuit of 1 bit magnitude comparator using only NAND gate. (CO1)

(b) Solve the following expression using Quine McCluskey method : (CO1)

$$Y = \Sigma m(0, 2, 6, 7, 8, 9, 10, 13, 15)$$

(c) Design 32 : 1 multiplexer using 8 : 1 or lower order multiplexer. (CO1)

2. (a) Design the circuit of BCD to GRAY code converter. (CO2)

(b) Implement the following circuit using PLA : (CO2)

$$Y_1 = \Sigma m(1, 2, 3, 5, 7)$$

$$Y_2 = \Sigma m(1, 3, 5, 6)$$

$$Y_3 = \Sigma m(2, 4, 5, 6)$$

P. T. O.

(c) What is Multiplexer ? Show how the following Boolean function may be implemented using a 4 : 1 multiplexer : (CO2)

$$Y(A, B, C) = \Sigma m(1, 3, 5, 7)$$

3. (a) What is race around condition ? Realize T flip-flop using D flip-flop. (CO3)

(b) Explain working of universal shift register with suitable diagram. (CO3)

(c) Design a synchronous counter for sequence : (CO3)

$$0 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 0$$

4. (a) Design MOD 13 Asynchronous counter. (CO4, CO6)

(b) An asynchronous sequential logic circuit is described by the following excitation and output function : (CO4, CO6)

$$Y = (X_1 \text{ XOR } X_2) + (X_1 \text{ OR } X_2)y$$

$$Z = y$$

(i) Draw the logic diagram of circuit.

(ii) Derive the transition table and output map.

(c) Design and explain the circuit of 4-bit PISO shift register. (CO4, CO6)

5. (a) Explain with neat diagrams TTL NAND gate. (CO5, CO6)

(b) Explain operation of ECL OR gate. (CO5, CO6)

(c) How are logic families classified ? Explain various characteristics of logic families. (CO5, CO6)

Roll No. ....

TEC–303

B. TECH. (ECE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

NETWORK ANALYSIS AND SYNTHESIS

Time : Three Hours

Maximum Marks : 100

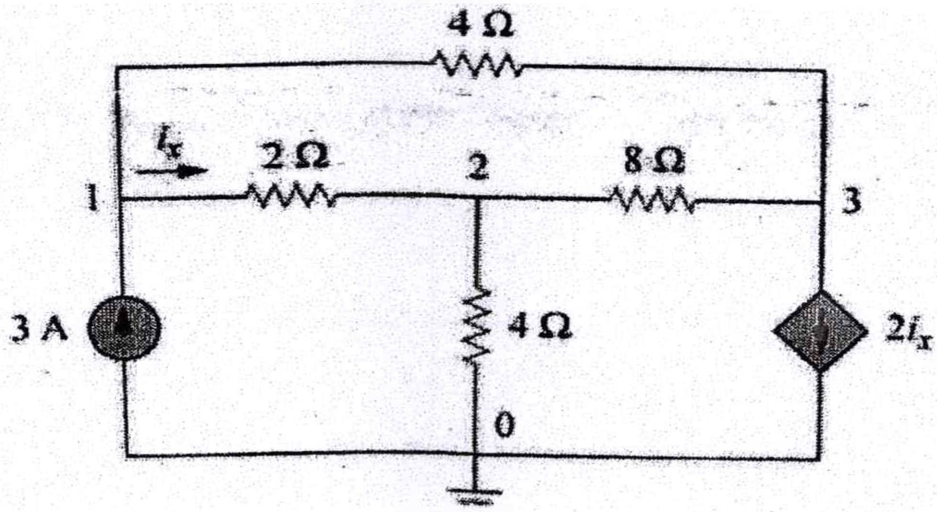
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

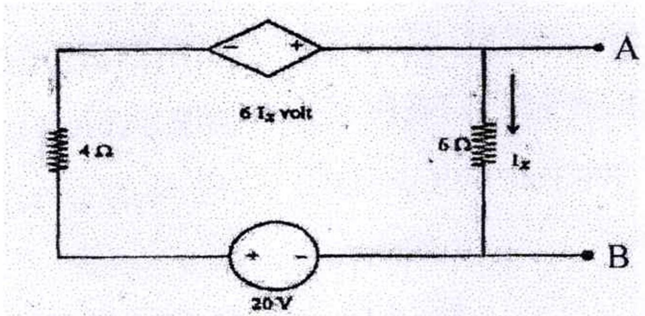
1. (a) Identify the node voltages in the circuit shown in the figure : (CO1)



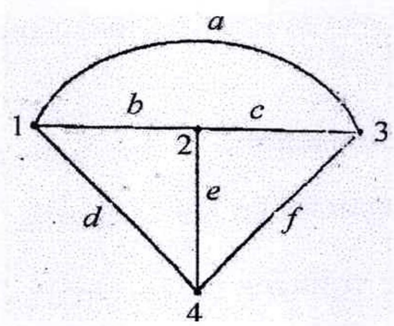
P. T. O.

(b) State and prove Maximum Power Transfer theorem with example. (CO1)

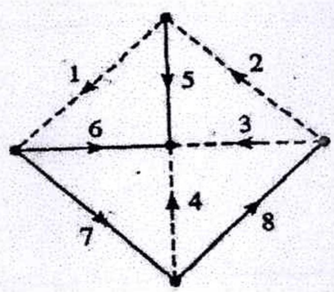
(c) Find the Thevenin's and Norton's equivalent circuit for the network shown in figure below : (CO1)



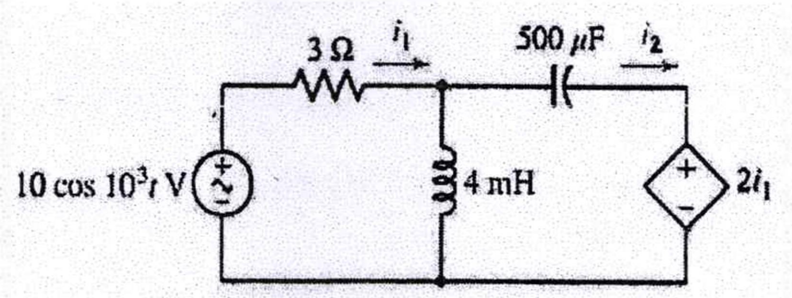
2. (a) Write fundamental cut set matrix and fundamental tie set matrix of any tree of one of the graph shown in figure below : (CO2)



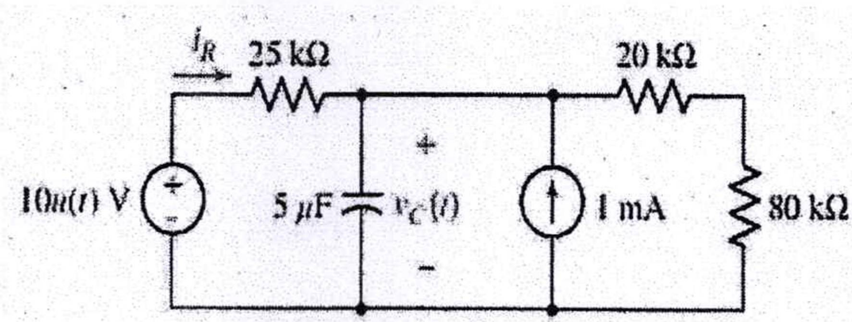
(b) Develop Tie set matrix and determine branch currents in term of loop current. (CO2)



(c) Obtain expressions for the time-domain currents i_1 and i_2 in the circuit given below : (CO2)



3. (a) Find $v_c(t)$ at t equal to (i) 0–; (ii) 0 +; (iii) ∞; (iv) 0.8 s. (CO3)



(b) Derive equation for transient response in parallel RLC circuit and hence deduce the condition for over damped and under damped case. (CO3)

(c) A series resonant circuit is composed of the elements R = 5Ω, L = 20 mH, and C = 1 mF. Compute : (CO3)

(i) Resonant frequency (ω_0)

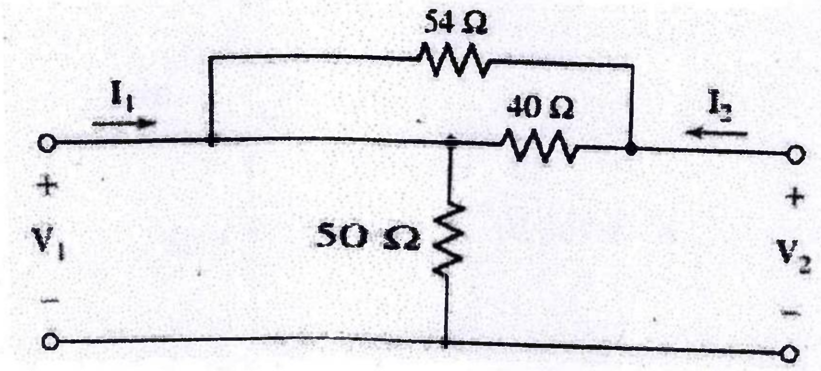
(ii) Quality Factor (Q_0)

(iii) Bandwidth

(iv) α

(v) β

4. (a) Obtain a complete set of y parameters to describe the two-port network : (CO4)



(b) Apply the concept of reciprocity to determine the condition on : (CO4)

(i) Y-parameters

(ii) T-parameters

(c) Given : (CO4)

$$t = \begin{bmatrix} 3.2 & 8\Omega \\ 0.2\text{S} & 4 \end{bmatrix}$$

find

(i) z

(ii) t for two identical networks in cascade

(iii) z for two identical networks in cascade

5. (a) Explain transfer function, poles, zeroes, pole-zero plot, characteristic equation, type, order of a system with the help of an example. (CO5, CO6)

(b) Realize the given impedance function in Foster form : (CO5, CO6)

$$F(s) = \frac{s(3s+8)}{(s+1)(s+3)}$$

(c) Explain the properties of positive real function. Determine whether the given function is positive real function or not : (CO5, CO6)

$$F(s) = \frac{(s+1)(s+2)}{(s+3)(s+4)}$$

Roll No. ....

TEC–304

B. TECH. (ECE) (THIRD SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

SIGNALS AND SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Find even and odd components of the following signals : (CO1)

(i) $x(t) = \sin t + \cos t + \sin t * \cos t$

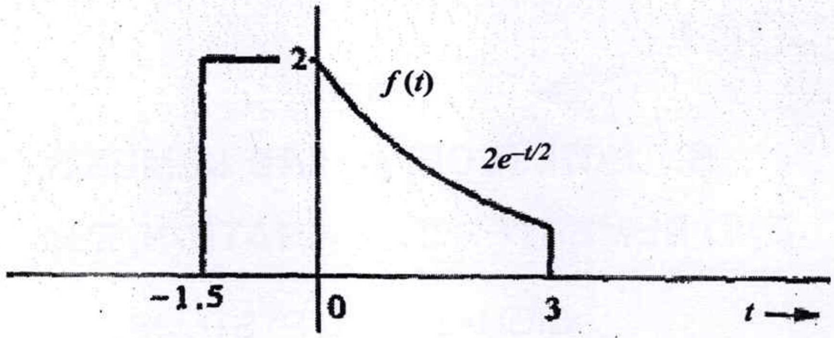
(ii) $x[n] = \{1, 0, -1, 2, 3\}$

(b) Show that : (CO1)

$$\delta(at) = \frac{1}{a}\delta(t)$$

P. T. O.

(c) Figure shows a signal $f(t)$, sketch and describe mathematically this signal time compressed by a factor 3. Repeat the problem for the same signal time expanded by factor 2. (CO1)



2. (a) Verify the associative property, that is : (CO2)

$$\{x[n] * h_1[n]\} * h_2[n] = x[n] * \{h_1[n] * h_2[n]\}$$

(b) Check whether the given system $y(t) = \text{even}\{x(t)\}$ is : (CO2)

(i) static or dynamic

(ii) linear or non-linear

(iii) causal or non-causal

(iv) time invariant or time variant

(c) Compute : (CO2)

$$y[n] = x[n] * h[n]$$

where $x[n] = \alpha^n u\,[n], h[n] = \beta^n u[n]$.

3. (a) Verify Parseval's theorem for the Fourier series. (CO3)

(b) Determine the complex exponential Fourier series representation for each of the following signals : (CO3)

(i) $x(t) = \cos(4t) + \sin(6t)$

(ii) $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$

(c) Derive the relation between trigonometric Fourier series and complex exponential Fourier series. (CO3)

4. (a) Find the Fourier transform of the following signals : (CO4)

(i) $x(t) = \text{sgn}(t)$

(ii) $x(t) = u(t)$

(iii) $x(t) = e^{-|t|}$

(b) State and prove time convolution property of Fourier transform. (CO4)

(c) State and prove frequency differentiation property of Fourier transform. (CO4)

5. (a) Find Laplace transform of the following signals : (CO5 and CO6)

(i) $x(t) = e^{-2t}\sin(2t)u(t)$

(ii) $x(t) = -t\,e^{-2t}u(t)$

(b) (i) Determine the initial value of $x(t)$ if its Laplace transform $X(s)$ is given by : (CO5 and CO6)

$$X(s) = \frac{(-2)e^{-5s}}{s(s+2)}$$

(ii) Determine the final value of $x(t)$ if its Laplace transform $X(s)$ is given by : (CO5 and CO6)

$$X(s) = \frac{1}{s(s-2)}$$

(c) What is meant by ROC ? Derive the relation between Fourier transform and Laplace transform. (CO5 and CO6)